Public Economics (ECON 131)
Final Exam Math Setup

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Contents

Note: This document summarizes the procedures to solve each type of exercise we have worked on this half of the semester, please flag any mistakes you may see, and if anything contradicts the section notes, follow the section notes!

1 Savings Taxation

1.1 Setup

You are given:

1. A utility function on $C1, C2$: $U(C1, C2)$. For example: $U = log(C1) + log(C2)$
2. Income Y earned only in the first period.
3. A rate of return r.
4. A savings tax rate τ. This is a tax on rS, and not S.

1.2 Solving the exercises

Solving the intertemporal problem is equivalent to solving a problem with two goods. The particularity is that the relative price of consumption in period two is given by the (net of tax) interest rate.

The (relative) price of consumption in period 2 is:

1. $\frac{1}{1+r}$ without a tax.
2. $\frac{1}{1+r(1-\tau)}$ with a tax.

The main step to solve these problems is to correctly set up the budget constraint:

1. BC without tax: $C_2 = (Y - C_1)(1 + r)$
2. BC with tax: $C_2 = (Y - C_1)(1 + r(1 - \tau)))$

The BC with tax is obtained by simply subtracting τrS from the first BC.

To solve the exercise, all that is left is to substitute C_2 in the utility function using the BC and take the first order condition with respect to C_1. Once C_1 is solved for, it is straightforward to obtain $S = Y - C_1$ and C_2 from the BC.

2 Externalities

2.1 Setup

You are given:

1. A supply function $Q_S = ...P$ or an inverse supply function $P_S = ...Q$
2. A demand function $Q_D = ...P$ or an inverse demand function $P_D = ...Q$
3. A marginal damage MD in \$ per unit. This can either be a constant or a function of Q, the process to solve the exercise is identical.

2.2 Solving the exercises

In these type of problems it may be slightly easier to work with the inverse supply and demand functions (P as a function of Q), you can invert any demand and supply function by isolating P on one side of the equation. For example $Q_D = 300 - \frac{1}{2}P$ becomes $P_D = 600 - 2Q$

2.2.1 Market Equilibrium

This is exactly the same as in the tax incidence exercises we saw at the start of the semester. Set $Q_D = Q_S$ or $P_D = P_S$ and solve.

2.2.2 Social Optimum

1. Set $P_D = P_S + MD$ or $P_S = P_D - MD$ and solve for the equilibrium quantity and prices.
2. Remember that, at the social optimum, the demand price and the supply price will be different by the value of MD at the social optimum (see graphs in section 9 for more detail). There will always be just ONE equilibrium quantity.
3. If you are asked about a Pigouvian / corrective tax, it will therefore be equal to MD. If MD is a function of Q, it will be equal to MD at the social optimum.
4. Remember the deadweight loss in a graph will always be above the Social Marginal Benefit curve and below the Social Marginal Cost, from the optimal Q^* (on the left with negative externalities) to the market equilibrium Q^M. Graphically the triangle will always point to the social optimum Q^* (see graphs from section for more detail).

3 Public Goods

3.1 Setup

You are given:

1. The number of agents (usually just two, but it is possible to have a problem with three types of agents).
2. A utility function on private consumption x and public good consumption M per agent $U_i(x_i, M)$ (they can be identical across agents). For example $U_i = \frac{2}{3}\log(x_i) + \frac{1}{3}\log(M)$
3. Income Y_i for each agent.

3.2 Solving the exercises

3.2.1 Market Equilibrium

1. Substitute in the utility functions $x_i = Y_i - m_i$ (assuming both prices are equal to 1, adjust by the prices otherwise) and $M = m_1 + m_2$ (assuming there are two agents).
2. Take the FOCs of the two utility functions with respect to m_i. This will give you the best response functions (m_1 as a function of m_2 and vice-versa).

3. Solve for the two choices of m_i by plugging in one best response function into the other. Note that if the two individuals are identical in utility and income, you can directly set $m_1 = m_2$ (see section for details).

3.2.2 Social Optimum

There are two ways of doing this. With the social utility function:

1. Set up the social utility function by adding up U_1 and U_2. Then solve exactly like in the market equilibrium case.
2. Substitute exactly like you did above.
3. Take the FOCs with respect to m_1 and m_2.
4. Obtain the optimal $M = m_1 + m_2$ by solving the best response functions.

Or use $MRS_1 + MRS_2 = MRT$:

1. $MRT = \frac{p_M}{p_x}$, assuming both prices are equal to 1, $MRT = 1$
2. $MRS_i = \frac{MU_{i,M}}{MU_{i,x_i}}$. Where $MU_{i,M} = \frac{\partial U_i}{\partial M}$ and $MU_{i,X} = \frac{\partial U_i}{\partial x_i}$. Remember not to substitute for x_i until after you have taken the derivative!
3. Solve for the socially optimal M.

4 Insurance

4.1 Setup

You are given:

1. Income in the good and bad states Y_G, Y_B. For example, in section, these were $Y_G = 500$, $Y_B = 10$. You can equivalently get Y_G and $d = Y_G - Y_B$, then $d = 490$ in our example.
2. Probability q of the bad state.
3. Utility as a function of consumption c, $U(C)$. For example $U = \log(c)$ or $U = \sqrt{c}$. Note that for insurance to be desirable, you need concavity in the utility function (i.e. risk aversion).

4.2 Solving the exercises

4.2.1 Setting up the expected utility functions

Let us assume $U(c) = \log(c)$ for illustration.

1. Without insurance: $\mathrm{E}[U(c)] = (1-q)\log(Y_G) + q\log(Y_B)$
2. With insurance: $\mathrm{E}[U(c)] = (1-q)\log(Y_G - p) + q\log(Y_B - p + b)$

4.2.2 Finding optimal insurance level b

If you have:

1. Actuarially fair premiums: $p = qb$
2. A concave utility function
3. q does not change with b (no moral hazard).

Then full insurance i.e. $b = Y_G - Y_B = d$ is optimal.

4.2.3 Finding the insurance market equilibrium

If insurance companies can observe individual types, then $p = q_i b$, where q_i is the probability of the bad state for type i.

Otherwise, insurances will charge all types $p_{pool} = \bar{q}b$, where $\bar{q}$ is the average probability of the bad state among the types that get insured. To solve this equilibrium (pooling equilibrium):

1. Find each types willingness to pay for insurance p_{max} by setting the expected utility of being insured equal to the expected utility of being uninsured and solving for p_max. Using the example from section:

$$E[U_{insured}] = E[U_{uninsured}] \Rightarrow \log(500 - p_{max}) = (1 - q)\log(500) + q\log(10)$$
$$\Rightarrow p_{max} = 500 - 500^{1-q}10^q$$

2. Obtain p_{max} for each type (each q) and compare it with $p_{pool} = \bar{q}b$. If $p_{max} < p_{pool}$ for a given type, that type will exit the market.
3. Recalculate p_{pool} Only with the types that remain in the market, and repeat the comparison. If $p_{max} < p_{pool}$ for the new p_{pool} then that type will also exit the market, and so on.